COMPUTER SCIENCE AND SOCIETY

AGENDA

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UNIT 3

ECONOMIC IMPACT OF THE INFORMATION SOCIETY

STUDY GOALS

On completion of this unit, you will be able to...

- define the economic significance of information society.
- explain how the mechanisms of globalization and monopolization have intensified in the information society.
- discuss the strengths and challenges of the Open movement.
- identify the effects of digitalization on the labor market.
- critically discuss the mechanisms for the protection of intellectual property.

EXPLAIN SIMPLY

1. Why do digital markets tend to develop **monopolies**?

2. Why is it hard to establish a rival to a service that has a **monopoly**?

3. What are potential advantages of **Open Source** software?

- German market for IT, telecommunications and consumer electronics: **€234.8 billion** in 2025
- new opportunities for access to **information, communication, economic activity**...
- … what about the **risks?**

https://www.bitkom.org/Bitkom-Dataverse/Marktzahlen?tab_name=ITK+Marktzahlen

- **Globalization**: process of exchanging material, technology, services, information and investments all across the globe.
- History:
  - Chinese **silk road**: from 2nd century BCE
  - 17th century: British **East India Company** owns more lands than the British crown
  - From 1940s: greater global interconnectedness, political alliances, growing trade, favourable global transportation.

Monopolies

- detrimental to the market: monopolist can charge **excessive prices** and is **less motivated** to develop its products or services
- institutions to ensure competition and prevent monopolies, e.g. Federal **Cartel Office** in Germany
- **Anti-trust measures**, e.g.
  - American Telephone and Telegraph Company (**AT&T**) split into 7 regional companies in 1984
  - **Microsoft**: repeatedly fined by European antitrust authorities

In the digital world

- **Network effect**: more members generate even more members
- **First mover advantage**: early players profit over proportionally
- **Zero marginal cost**: easy to scale quickly (even more so with **cloud computing**)

- Few large companies hold substantial share of their markets (e.g. Google, Apple, Facebook, Amazon)

- Early spirit of the computing scene: **sharing knowledge**
- Wikipedia: created and maintained by volunteers – **interest of the commons**
- Linux: free**, open source operating system** (anybody can edit). Used broadly, e.g. on Web Servers
- Many others, e.g. **Firefox, Libre Office, Python, Shortcut**…

- Free, often **high-quality** software
- **Sharing** the source code: more transparency, probably more secure
- … but potential issues of **compatibility** with other software
- … who supports the software? Who is liable?

More in Koranne (2011)

Sourcing:

- Companies in industrialized nations can relocate (outsource) work to **developing countries**
- Creates **good jobs** there, **new prospects** for people
- jobs in the country of origin potentially lost
- People in industrialized countries **competing with cheaper labor** around the world
- **Cultural difficulties** in cooperation...

- Automation – **cost efficiency** – loss of jobs
- AI expert systems **replacing human labor**: white collar jobs at risk
- **Universal basic income**: solution or risk?
- Working from home: **is the office dead?**
- **Blurring boundaries** work/ private life …

- can be protected by **encryption**, use of **watermarks**, and use of **legal notices**
- **Problems**: different laws in different countries (e.g. copying software legal in some countries)/ protected content easy to distribute online

Can **software** be **patented**?

- patent law originally created for **technical inventions**
- computer-implemented invention is patentable if it makes a **technical contribution**
- **difficult to define** what a contribution is in this case

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SESSION 3

TRANSFER TASK

Is AI killing jobs?

> *“Contemporary AI systems are now becoming human-competitive at general tasks, and we must ask ourselves: (…) Should we automate away all the jobs, including the fulfilling ones?”* (Future of Life Institute, 2023)

The famous open letter discusses potential risks associated with **ChatGPT and similar tools**. On the other hand, a study by the International Labour Organization (ILO) found that “Generative AI is likely to **augment** rather than destroy jobs” (ILO, 2023). What do you think?

1. In your group, form two teams.
   - **Team 1** will argue that Generative AI/ ChatGPT **poses a huge risk** to job markets and may lead to mass unemployment
   - **Team 2** will argue that Generative AI/ ChatGPT **does not pose such a risk**, and that there is no danger to job markets.
2. Spend some time **researching** arguments for your case, starting with the **resources** given above (see previous slide)!
3. Each team should **present** their point of view to the others in about **10 minutes**.
4. Which side did you find more convincing?

TRANSFER TASK
PRESENTATION OF THE RESULTS

Please present your results.
The results will be discussed in plenary.

LIST OF SOURCES

Future of Life Institute. (2023). Pause Giant AI Experiments: An Open Letter. Future of Life Institute. https://futureoflife.org/open-letter/pause-giant-ai-experiments/

ILO. (2023, August 21). *Generative AI likely to augment rather than destroy jobs* [Press release]. http://www.ilo.org/global/about-the-ilo/newsroom/news/WCMS_890740/lang--en/index.htm

Koranne, S. (2011). *Handbook of Open Source Tools*. Springer US. https://doi.org/10.1007/978-1-4419-7719-9